

Lecture-2: Foundation of Energy Conversion

Energy Conversion

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Theoretical Fundamentals

Phenomena of nature (Energy Exchange)

Interactions between 3 types of energy exchanges

- Magnetic Energy
- Electric Energy
- Mechanical Energy

3 systems/circuits

- Magnetic System/Circuit
- Electric System/Circuit
- Mechanical System

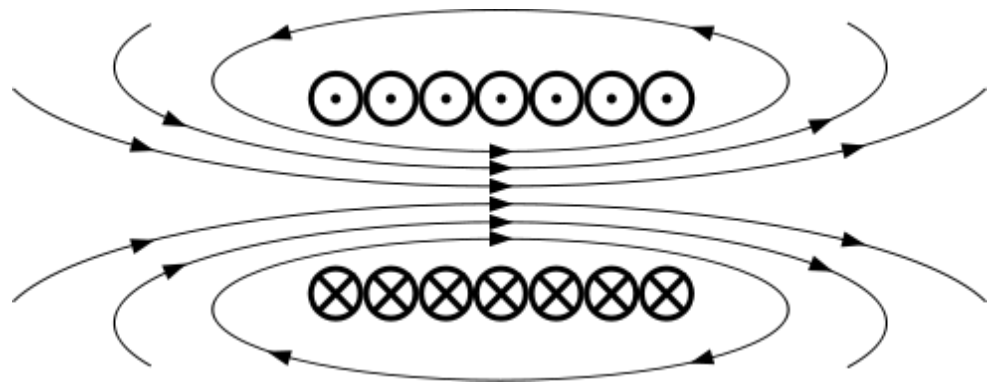
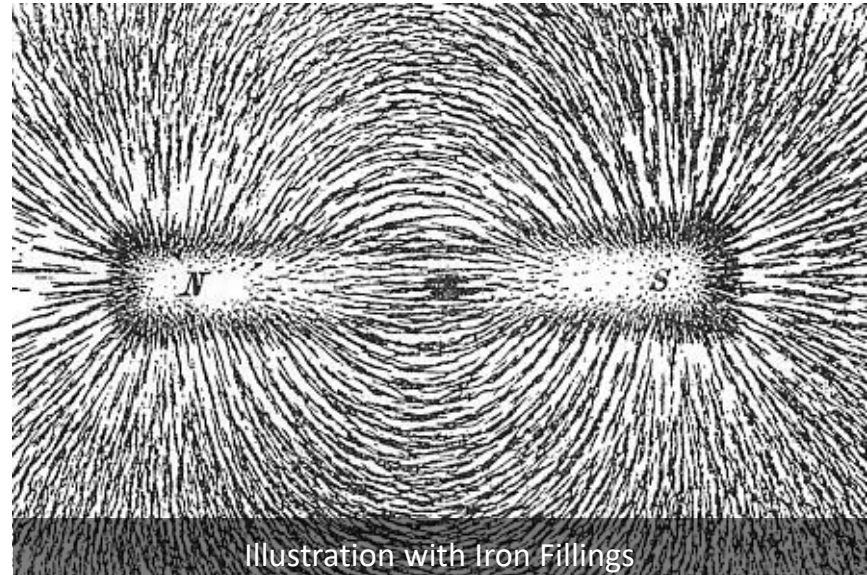
Overview of Physics

- Oersted's Discovery and Ampere's Law
- Magnetization
- Faraday-Lenz Law
- Motor Action: Current Carrying Conductor in Magnetic Field

Phenomenon of Nature I: Discovered by Hans Oersted from Denmark

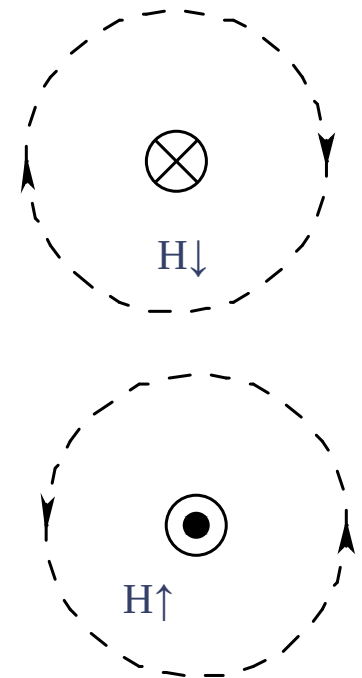
Electric field $\rightarrow$ electric
field/potential

Magnetic field $\rightarrow$ magnetic
field/potential



Definition/Description: Magnetic Field ($\vec{H}$)

- Important concept:
Magnetic field/intensity, H , produced by current carrying conductor
- Magnetic field/intensity/strength
 - Field: force in space/3D
 - Vector: magnitude & **direction (right-hand rule)**
 - Measure/unit: amperes per meter (A/m)



Direction of H

- Magnetic field/intensity, H , produced by current carrying conductor

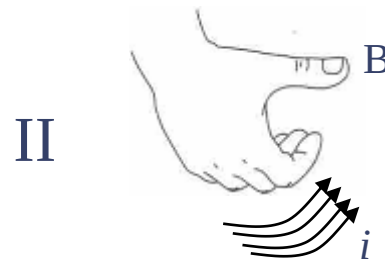
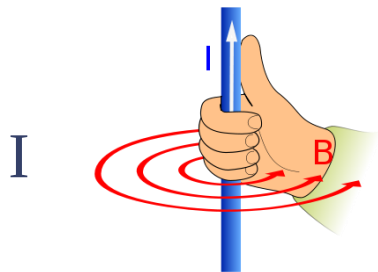
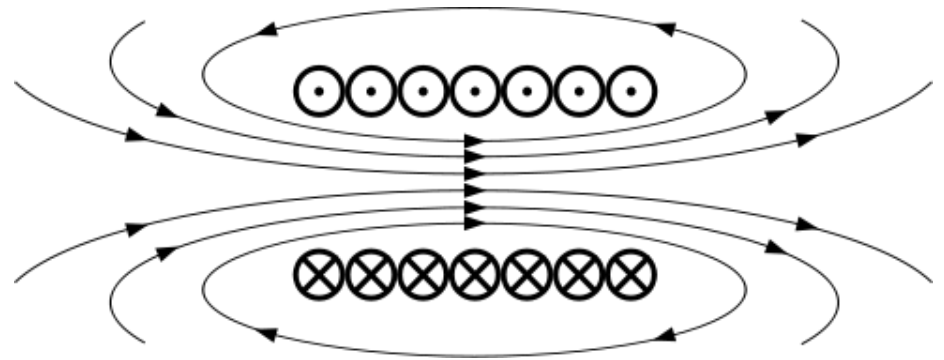
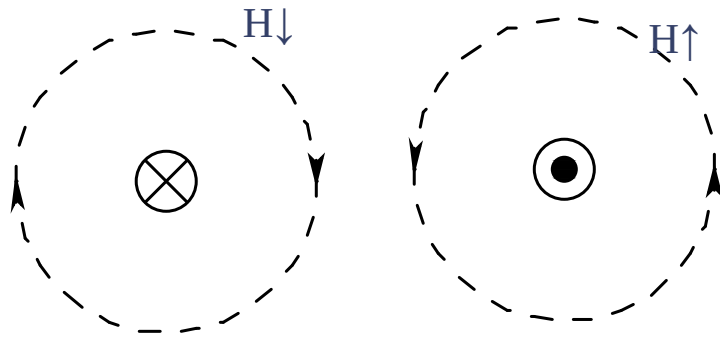
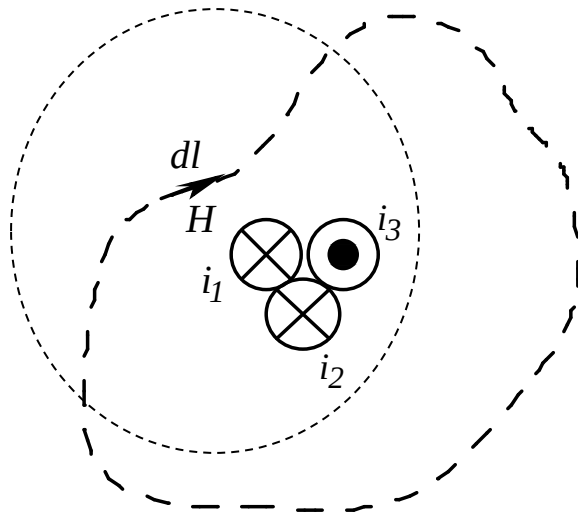


Illustration with Iron Fillings

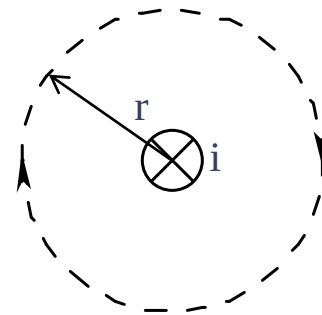
Quantification of Magnetic Field/Intensity ($\vec{H}$)

- Ampere's Law

$$\oint_{\text{closed path}} H d\ell = \sum i$$



Irregular shape



Regular shape: $\pi \cdot 2r \cdot H = i$

Let's Approximate

A value or quantity that is nearly but not exactly correct.

Equivalent is equivalent while **approximate** is approaching;
proximate; nearly resembling

In science and engineering, you often approximate complicated functions by simpler ones which are easier to calculate with, and which show the relations between the variables more clearly. Of course, the approximation must be close enough to give you reasonable accuracy.

Ampere's Law

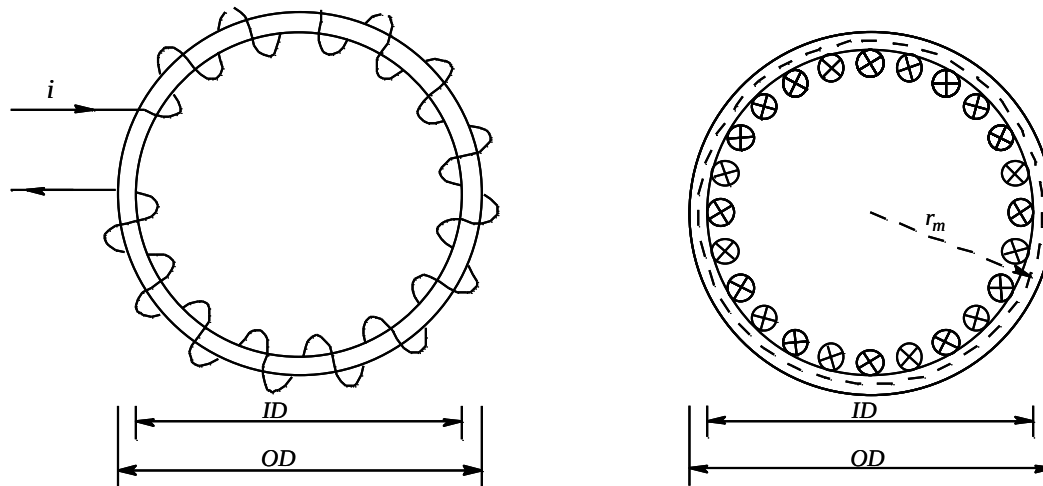
The line integral of H about any closed path is exactly equal to the net direct current enclosed by that path....

- H -(Magnetic Field)
- l_c : the mean path length of the core
- Ni : magnetomotive force (F), mmf, Ampere-turn (A-t)

$$\oint H \cdot dl = \sum I \text{ or } I_{Net}$$

For regular shape – l can be computed: $Hl_c = Ni$, $H = \frac{Ni}{l_c}$

Example (Symmetric Structure)



$$\text{Mean radius, } r_m = \frac{1}{2} \left(\frac{ID + OD}{2} \right)$$

$$l_m = 2\pi r_m$$

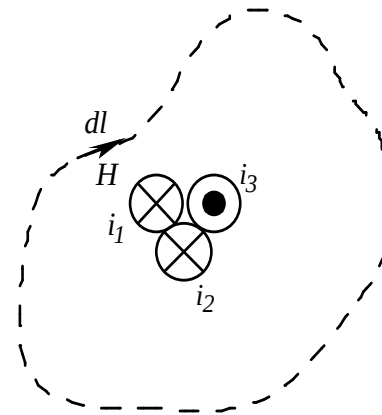
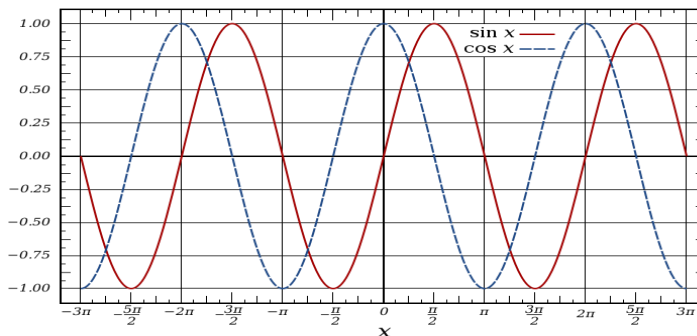
$$\text{Ampere's Law} \Rightarrow H_m = \frac{Ni}{2\pi r_m} = \frac{Ni}{l_m}$$

Further discussion: changing current produces changing magnetic field/force

- Change in current direction
- Change in current magnitude
- Change in both direction and magnitude
- Change in density



$$\oint_{\text{closed path}} H d\ell = \sum i$$



Phenomenon of Nature II: Magnetization & Ferromagnetic Material

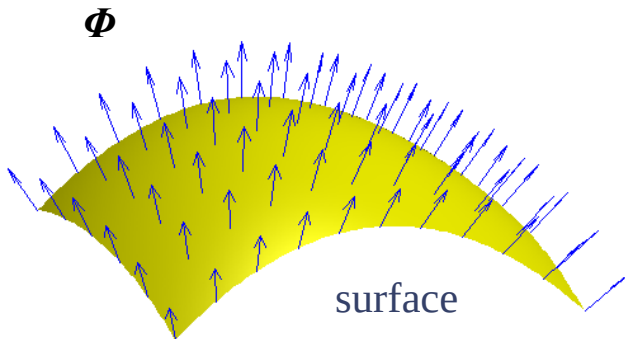
Magnetic Flux (Φ) and Flux Density (B)

Φ : measure of total quantity of magnetism strength pass through a surface (Weber)

Φ (flux) is closed, like current

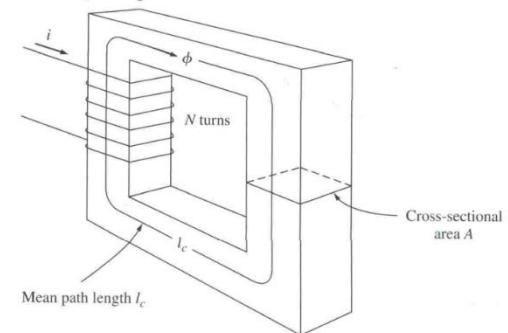
B: Magnetic flux density is the amount of magnetic flux per unit area of a section, perpendicular to the direction of flux (Weber/mm)

Φ and B: $B = \Phi / A$, $\Phi = BA$ or



$$\Phi_m = \iint_S \mathbf{B} \cdot d\mathbf{S}$$

Figure 1-3 | A simple magnetic core.

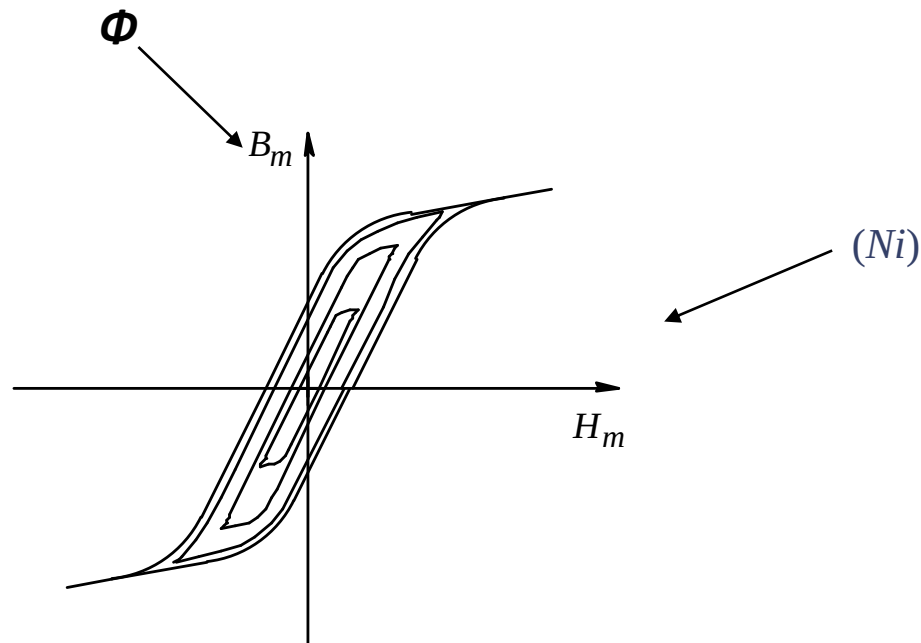


Two Important Effects in Magnetism

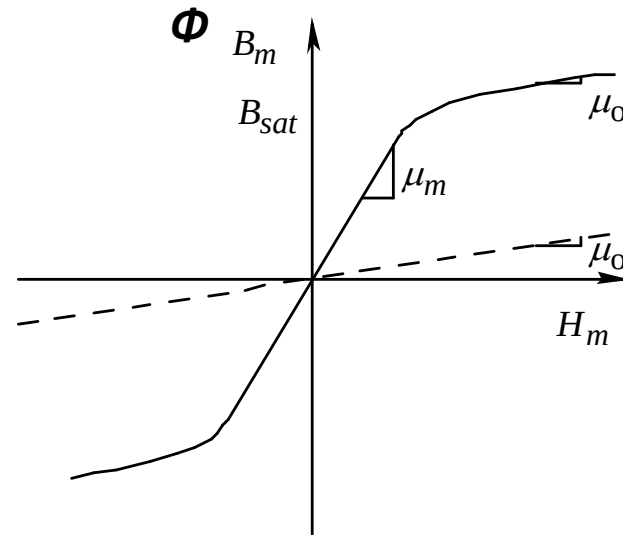
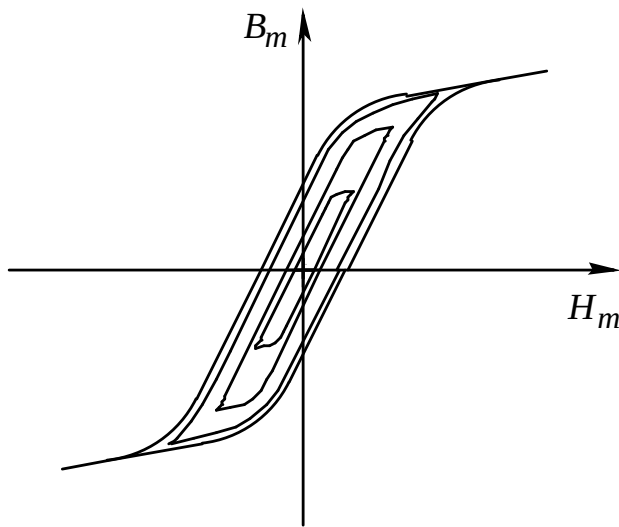
Saturation effect

Hysteresis effect

- Coercivity
- Remanence



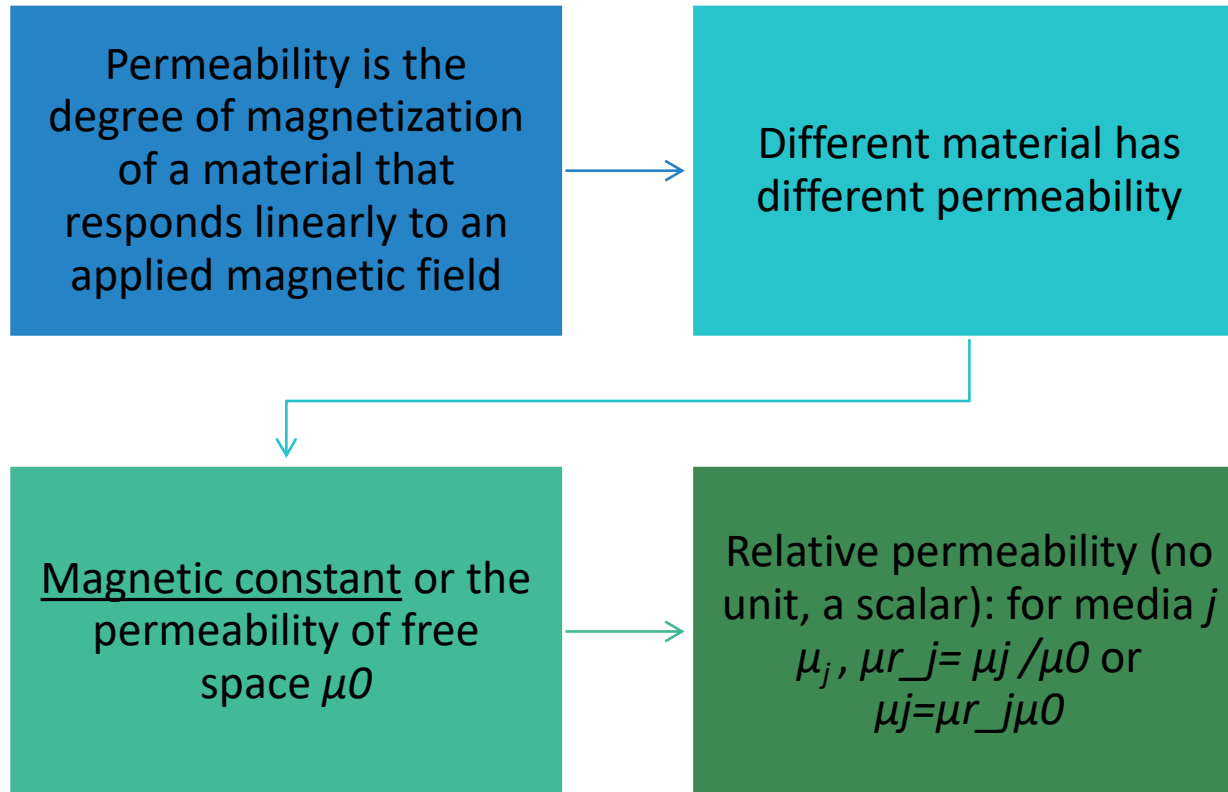
Magnetization of Ferromagnetic Media



Linear approximation
In Saturation

$$B_m = \mu_m H_m$$

Permeability



Magnetization of Ferromagnetic Media

Resistance to the magnetization?

- Vacuum
- Air
- Wood
- Iron
- Ferromagnetic material (least resistance, very easy to be magnetized)

Measured by μ (permeability) -
different material has different μ , or

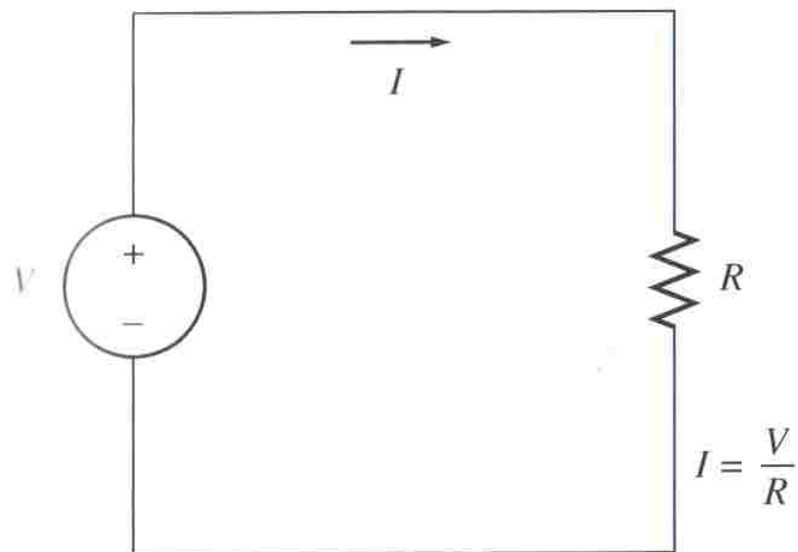
by relative permeability : $\mu_j = \mu_r \mu_0$

Conductivity

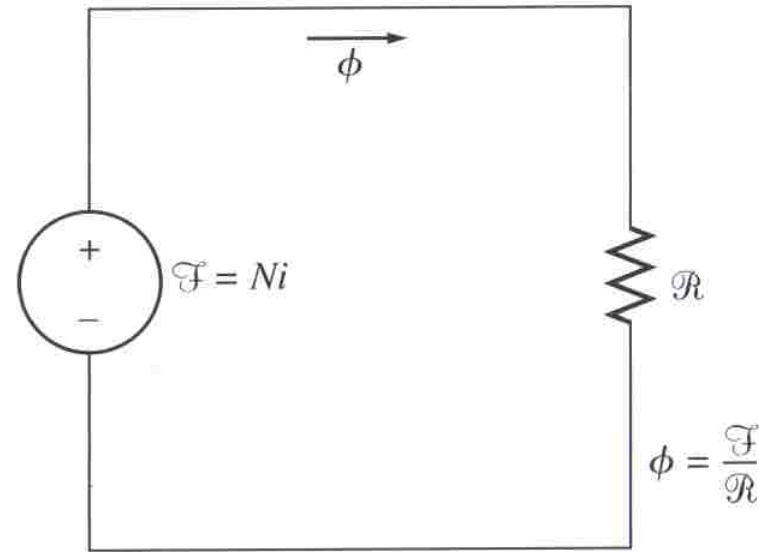
Permeability

Iron	excellent	excellent
Al, Copper	excellent	poor
Wood	poor	poor

Conductivity and Permeability



(a)



(b)

Introduction to Magnetic Circuit

Magnetic Circuits

$$\mathfrak{F} = \phi \mathfrak{R}$$

$$\phi = BA = \frac{\mu NiA}{l_c} = Ni \left(\frac{\mu A}{l_c} \right) = \mathfrak{F} \left(\frac{\mu A}{l_c} \right) = \mathfrak{F} P = \frac{\mathfrak{F}}{\mathfrak{R}}$$

$$\mathfrak{R} = \frac{l_c}{\mu A}$$



Magnetic Circuits – *cont.*

$$\mathfrak{R}_1 = \ell_1 / \mu A_1, \mathfrak{R}_2 = \ell_2 / \mu A_2, \mathfrak{R}_3 = \ell_3 / \mu A_3$$

- **For a number of reluctances in series :**

$$\mathfrak{R}_{eq} = \mathfrak{R}_1 + \mathfrak{R}_2 + \mathfrak{R}_3$$



- **For a number of reluctances in parallel :**

$$\frac{1}{\mathfrak{R}_{eq}} = \frac{1}{\mathfrak{R}_1} + \frac{1}{\mathfrak{R}_2} + \frac{1}{\mathfrak{R}_3}$$



Magnetic Circuits

Summary:

NAME	SYMBOL / UNITS	EQUATION
Magneto-motive Force (mmf)	F (Ampere-turns) or (A-t)	$F = Ni$
Magnetic Field Intensity	H (A-t/meter)	$H = Ni/l$ (path length)
Flux Density	B (Tesla) 1 Tesla = 1 Wb/meter	$B = \mu H$
Permeability	μ	$\mu = \mu_r \mu_0$ μ_r relative permeability μ_0 free space permeability
Flux	Φ (Webers, Wb)	$\Phi = BA$ (crosssectional area in square meters)
Flux Linkage	λ (Weber-turns) or Wb-t	$\lambda = N \Phi$
Reluctance	$\mathcal{R}$ (A-t/Wb)	$\mathcal{R} = l/\mu = \mu_r \mu_0 A$
Inductance	L (Henries)	$L = \lambda/i = N^2/\mathcal{R}$

- Magnetic circuit model is often used in the design of electric machines to simplify the otherwise quite complex design process.
 - Ohm's law in an electric circuit $\rightarrow V = RI$
 - Hopkinson's law in the magnetic circuit is $\rightarrow \mathfrak{S} = \phi \mathcal{R}$

Electric Circuit		Magnetic Circuit	
Current	I (Amps)	Flux	Φ (Wb)
emf	V (Volts)	mmf	F (A-t)
Resistance	R (Ohms)	Reluctance	$\mathcal{R}$ (A-t/Wb)
Conductivity	σ (S/m)	Permeability	μ (H/m)

Summary

I (electric circuit/field) → a
fundamental interaction of matter →
H (magnetic field/intensity/strength)
→ magnetize the media/medias →
 ϕ (magnetic flux)

Changing magnetic field will induce
voltage potential of a circuit
(conductor)

Concepts

- ϕ and B
- Two characteristics of ferromagnetic material
- permeability of material

Force will be induced on a current
carrying conductor if it's placed in a
magnetic field.

Phenomenon of Nature III:

Changing Magnetic Flux Linkage Induces Electric Field
Described by Faraday's Law and Lenz's Law

Phenomenon III

If changing flux passes through a turn of a coil or wire ring, a voltage will be induced,

$$e_{ind} = -\frac{d\phi}{dt} \quad , \quad e_{ind} = -N \frac{d\phi}{dt} \text{ (for } N \text{ turns)}$$

Faraday's Law (magnitude): the voltage magnitude is directly proportional to the rate of change in the flux with respect to time.

Lenz's law (the minus sign on RHS): the direction of the induced voltage in the coil is such that if the coil ends were short circuited, it would produce current that would cause a flux opposing the original flux **change**.

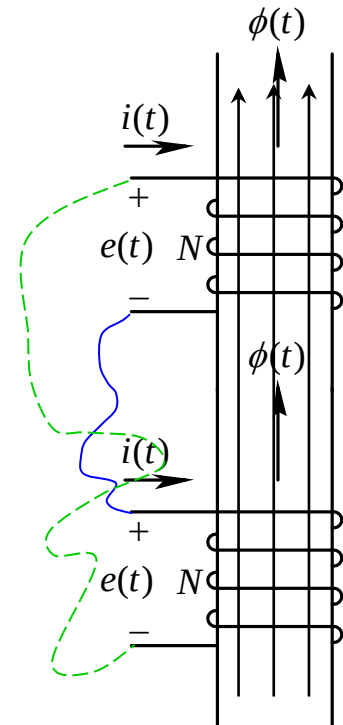
Flux Linkage: Multiple Turns

If there are N turns in the coil of wire, the total voltage on the coil is...

$$e_{ind} = \sum_{i=1}^N e_i = \sum_{i=1}^N \frac{-d(\phi_i)}{dt} = \frac{-d}{dt} \left(\sum_{i=1}^N \phi_i \right)$$

The flux linkage λ of the coil is defined as

$$\lambda = \sum_{i=1}^N \phi_i \quad (\text{weber-turns})$$



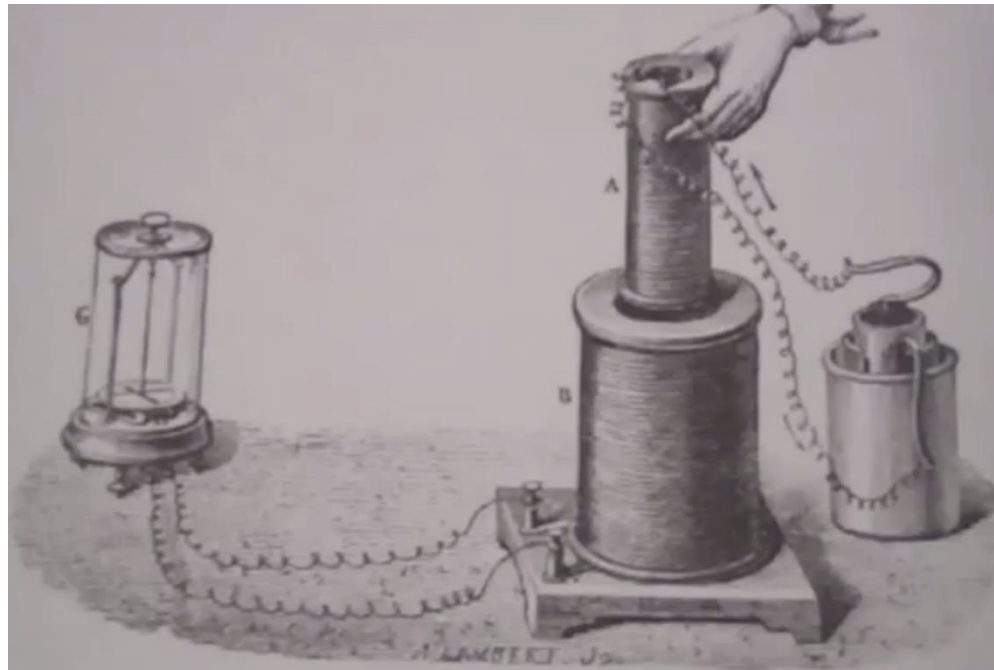
Induced Voltage in Electric Circuit

Changing magnetic **field** **Flux Linkage** induces electric potential in circuit, if close loop – product current

$$e_{ind} = -\frac{d\phi}{dt},$$

for N turns,

$$e_{ind} = -N \frac{d\phi}{dt}$$

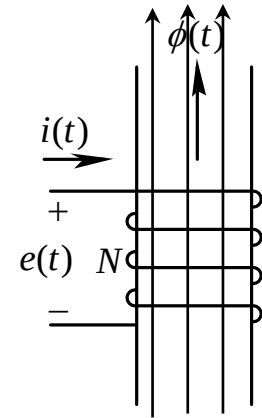


Summary of Voltage Induction

Faraday's Law: induced voltage $|e| = -\frac{d\lambda}{dt} = -N\frac{d\phi}{dt}$

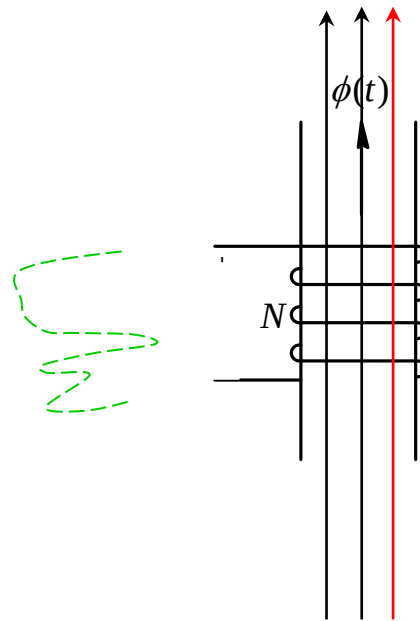
Lenz's Law: Polarity of induced voltage

- When current and flux directions are consistent
- Direction of flux induced by the imaginary current is determined using the change of flux based on Lenz's Law

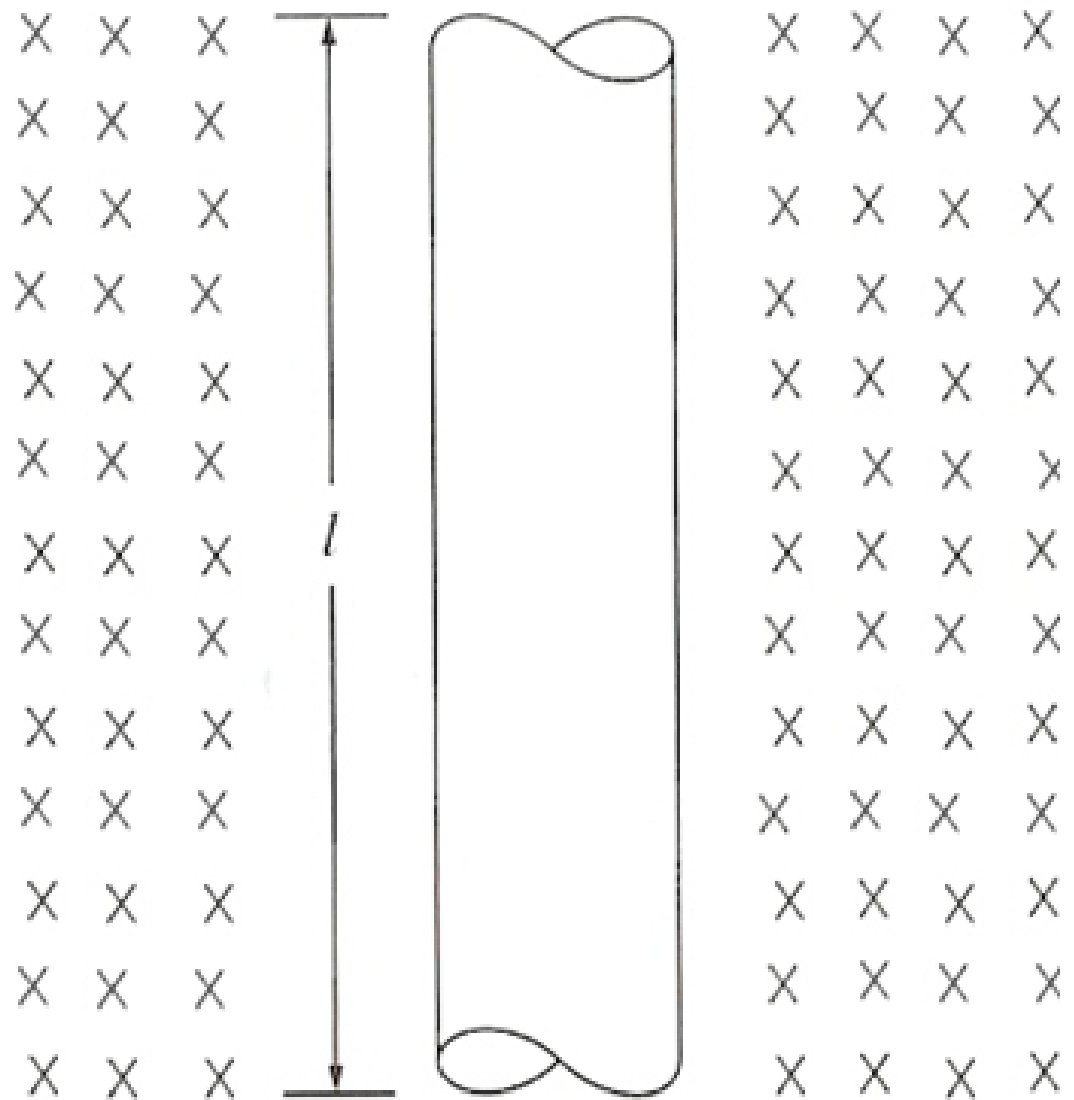


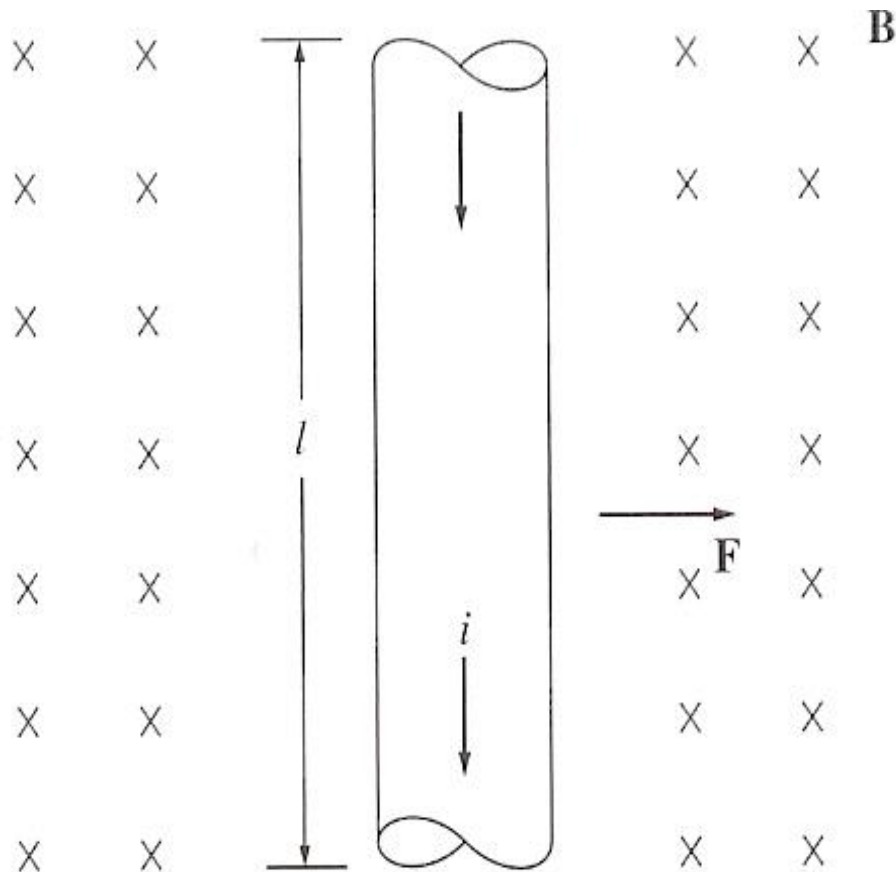
Change of flux

A way to figure out voltage polarity



Type I Change of Total Number of Flux





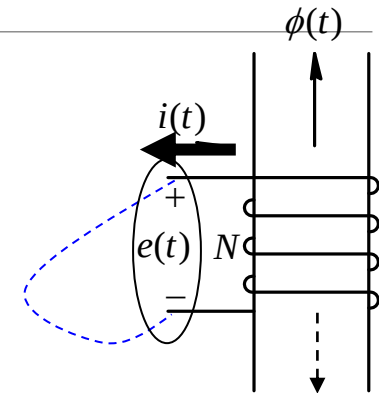
Type II Movement of Circuit

Example

Important formula for changing flux

$$\frac{d\phi}{dt} = \frac{\Delta\phi}{\Delta t} = \frac{\phi_{t+1} - \phi_t}{\Delta t}$$

$$\frac{d(A\sin(\omega t + \theta))}{dt} = A\omega \cos(\omega t + \theta)$$



If $N = 5$, $\phi = \sin(t)$, what is the magnitude of the induced potential in the circuit shown in the figure?

If $N = 5$, ϕ is increasing at a speed of 5 Weber in each five seconds, what is the magnitude of the induced potential in the same circuit, and what is the potential polarity?

Another mathematical description of Faraday's Law (Generator Action)

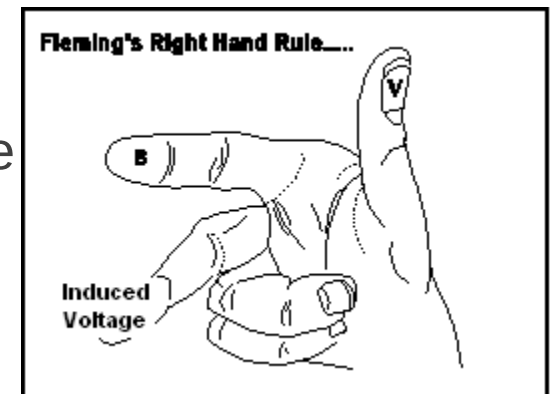
A wire which moves through a magnetic field induce a voltage in the wire.

The voltage induced in the wire is given by

$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l}$$

The voltage in the wire will be built up so that the positive end is in the direction of the vector

If the direction of the vector $\mathbf{l}$ is same as that of the vector $\mathbf{v} \times \mathbf{B}$, the voltage is given by Fleming's right-hand rule.



Phenomenon of Nature IV:

Induced Invisible Force on Current Carrying Conductor in Magnetic Field



Production of Induced Force on a Wire (Motor Action)

A magnetic field induces a force on a current-carrying wire within the field.

The force induced on the conductor is given by

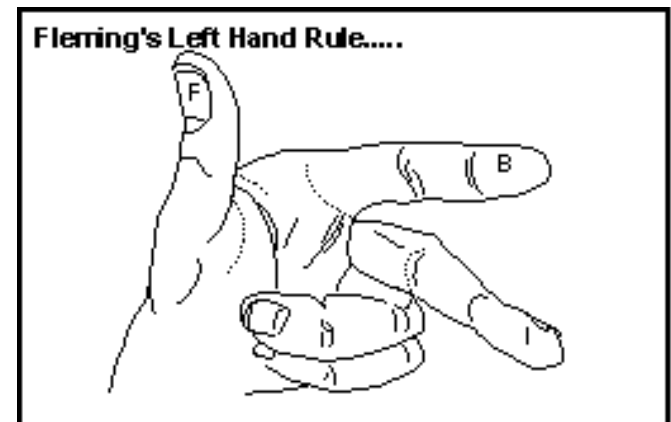
$$F = i(l \times B)$$

The direction of the force is given by Fleming's left-hand rule.

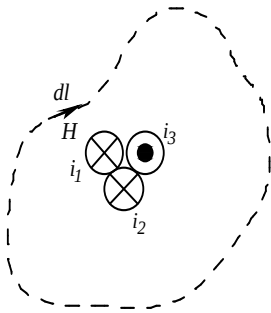
The magnitude of the force

is given by... $F = ilB \sin \theta$

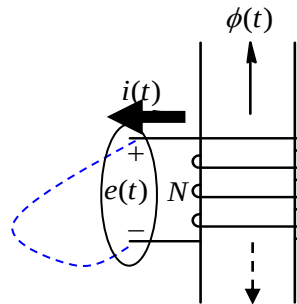
Example



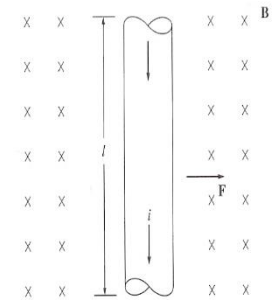
Summary



$$\oint H \cdot dl = \sum I \text{ or } I_{Net}$$



$$e_{ind} = -N \frac{d\phi}{dt}$$



$$\vec{F} = q\vec{v} \times \vec{B}$$

$$\vec{F} = I\vec{l} \times \vec{B}$$

Conductivity

Permeability

Iron	excellent	excellent
Al, Copper	excellent	poor
Wood	poor	poor

Procedure: from i to Φ

- **Steps**

- Find F ($\sum i$ or Ni)
- Find H and perimeter (Amperes law)
- Find B (B - H curve, or $B = \mu H = \mu_r \mu_0 H$)
- Find Φ

$$F = Ni$$

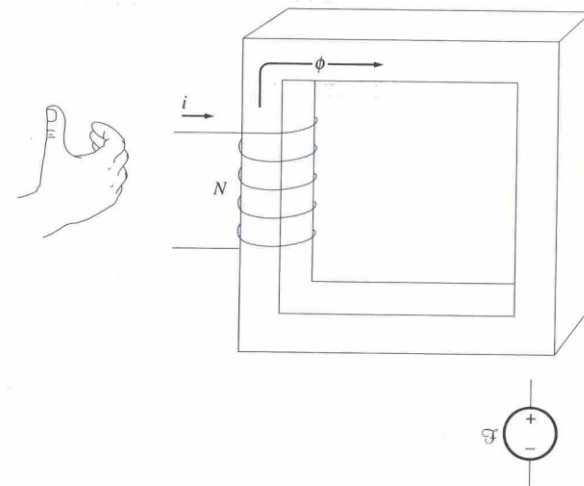
$$Hl = F = Ni$$

Direction of H

$$B = \mu H = \mu_r \mu_0 H$$

$$\Phi = \int_s B ds = BS$$

**Establish
Equations,
solve for
Unknowns**

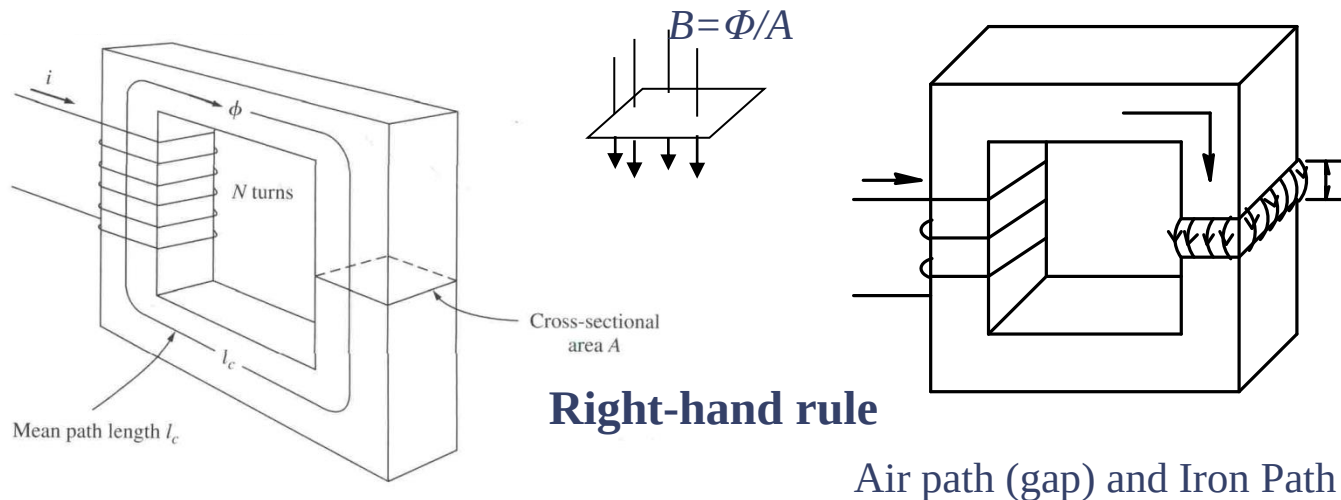


Magnetization of Multiple Medias

Flux is closed and

$$\phi_m = \phi_g = \phi$$

Most of flux will be formulated in the permeability with the least difficulty



Example: Multiple Magnetic Medias

Example: A Core with Air Gaps

- Equal flux along the path
- Various intensity and density
- Driven by total MMF

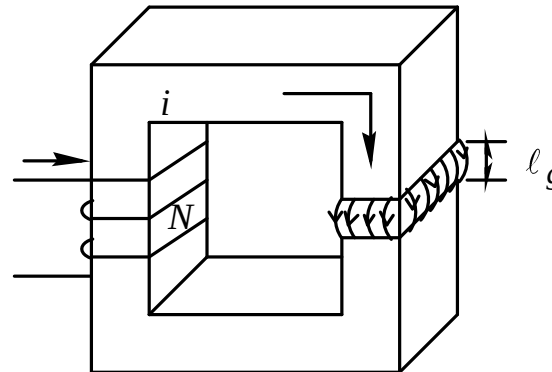
! $\underline{H_m \ell_m + H_g \ell_g = Ni} \quad F = Ni$

! $\underline{B_m = \mu_m H_m, \quad B_g = \mu_o H_g}$

$$\frac{B_m}{\mu_m} \ell_m + \frac{B_g}{\mu_o} \ell_g = Ni$$

! $\underline{\phi = A_m B_m = A_g B_g} \quad \boxed{\phi_m = \phi_g = \phi}$

$$B_m = \frac{\phi}{A_m} \quad B_g = \frac{\phi}{A_g}$$



$$\phi_m \left(\underbrace{\frac{\ell_m}{A_m \mu_m}}_{\mathfrak{R}_m} + \underbrace{\frac{\ell_g}{A_g \mu_o}}_{\mathfrak{R}_g} \right) = Ni \quad \phi_m = \frac{F}{\mathfrak{R}}$$

analogous to $I = \frac{V}{R} \quad I \left(\underbrace{\frac{l_1}{A_1/\rho_1}}_{R_1} + \underbrace{\frac{l_2}{A_2/\rho_2}}_{R_2} \right) = V$